

AMAZON / E-COMMERCE SALES ANALYTICS

Data Analytics & Business Analytics Project Report

Prepared By	Suresh Kumar
Tools	Excel, SQL, Power BI, Data Visualization
Dataset	ECOMM DATA.xlsx
Analysis Period	01 Jan 2011 - 31 Dec 2014
Primary Objective	Convert transactional data into actionable business insights

Portfolio-ready project: This report documents the analytical workflow, dashboard design, KPIs, business findings, and recommendations derived from the supplied e-commerce order dataset.

1. Executive Summary

The project analyzes 51,290 order-line records covering four years of e-commerce transactions. The dataset generates \$12.64M in sales, \$1.47M in profit, and 178.3K units sold. The overall profit margin is 11.61%. The analysis combines data preparation, KPI analysis, segmentation, geographic analysis, product/category profitability, shipping analysis, and interactive Power BI visualization.

The strongest category by revenue is Technology (\$4.74M), while the strongest market is APAC (\$3.59M). Standard Class is the dominant shipping mode with \$7.58M in sales. At the sub-category level, Tables is the major profitability concern, producing a loss of \$-64.1K.

Sales increased from \$2.26M in 2011 to \$4.30M in 2014, a rise of approximately 90.3%. This indicates sustained business expansion, while the category and sub-category analysis highlights where margin improvement should be prioritized.

2. Project Objectives

- Measure overall sales, profit, quantity, orders, customers, and shipping performance.
- Identify high-performing and low-performing categories, sub-categories, markets, countries, and states.
- Analyze yearly growth and profitability trends from 2011-2014.
- Evaluate customer segments and shipping modes for business planning.
- Use Power BI slicers and visuals to enable category-level drill-down and interactive analysis.
- Translate analytical findings into practical business recommendations.

3. Dataset Overview

The supplied Excel workbook contains three sheets: Orders, Returns, and People. The Orders table is the primary analytical fact table, while Returns supports return-rate analysis and People provides regional ownership context.

Sheet	Rows	Structure	Primary Use
Orders	51,290	24 columns	Sales, profit, quantity, customer, geography, product, shipping
Returns	1,175	3 columns	Returned order IDs and market
People	13	2 columns	Regional person/manager mapping

Important fields used in the analysis include Order Date, Ship Date, Ship Mode, Customer ID, Segment, Country, Market, Region, Category, Sub-Category, Product Name, Sales, Quantity, Discount, Shipping Cost, Profit, and Order Priority.

4. Analytical Methodology

- Data ingestion: Loaded the Excel workbook and reviewed all three sheets.
- Data validation: Checked row counts, data types, dates, categorical fields, and numerical measures.
- Feature preparation: Derived Year from Order Date and calculated profitability ratios and return rate.
- Aggregation: Grouped transactions by category, sub-category, market, country, state, segment, shipping mode, and year.

- KPI modeling: Calculated total sales, profit, quantity, orders, customers, margin, discount, and returns.
- Visualization: Built an interactive Power BI dashboard with KPI cards, slicers, bar charts, donut charts, and comparative views.
- Business interpretation: Converted descriptive findings into actions related to pricing, product mix, markets, logistics, and profitability.

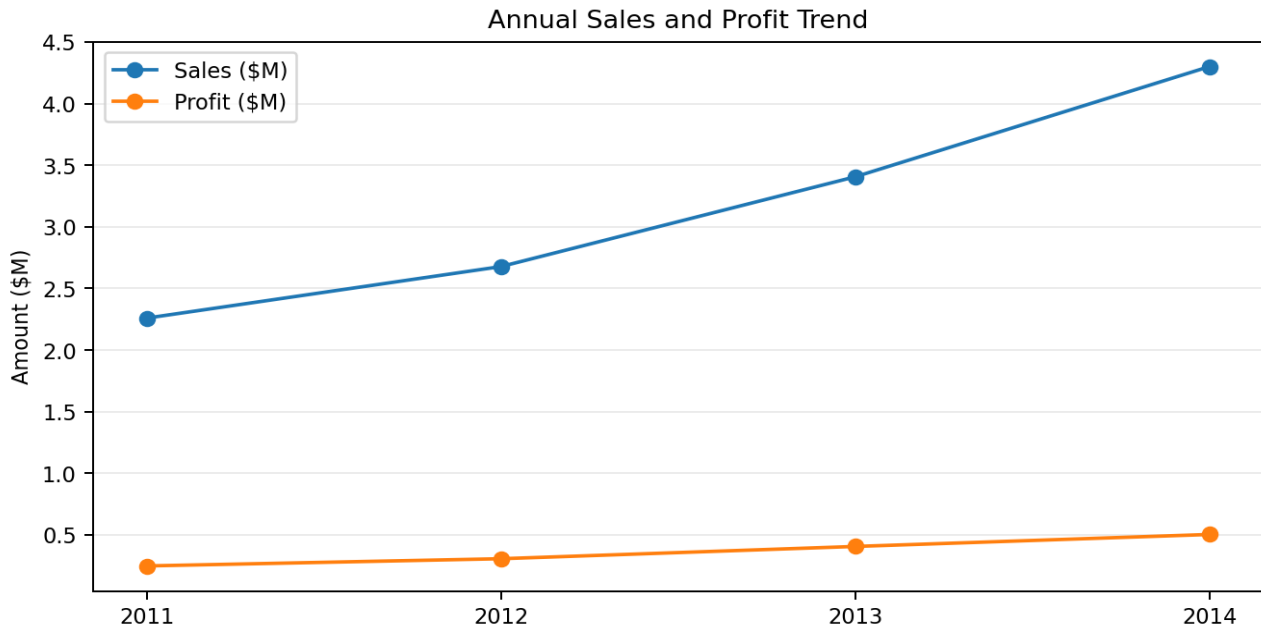
5. KPI Dashboard Summary

KPI	Value	Business Meaning
Total Sales	\$12.64M	Overall revenue generated
Total Profit	\$1.47M	Overall profit; margin 11.61%
Total Quantity	178.3K	Units sold across all transactions
Orders	25,035	Distinct customer orders
Customers	1,590	Distinct customers
Markets	7	Geographic markets covered
Countries	147	Countries represented
Return Rate	4.69%	Returned orders / distinct orders
Weighted Discount	0.11%	Sales-weighted average discount

The Power BI dashboard uses a Category slicer to compare Furniture, Office Supplies, and Technology. The supplied screenshots demonstrate how the dashboard dynamically updates KPI cards, country/state rankings, market mix, category contribution, and shipping-mode performance when a category is selected.

6. Sales & Profitability Trend Analysis

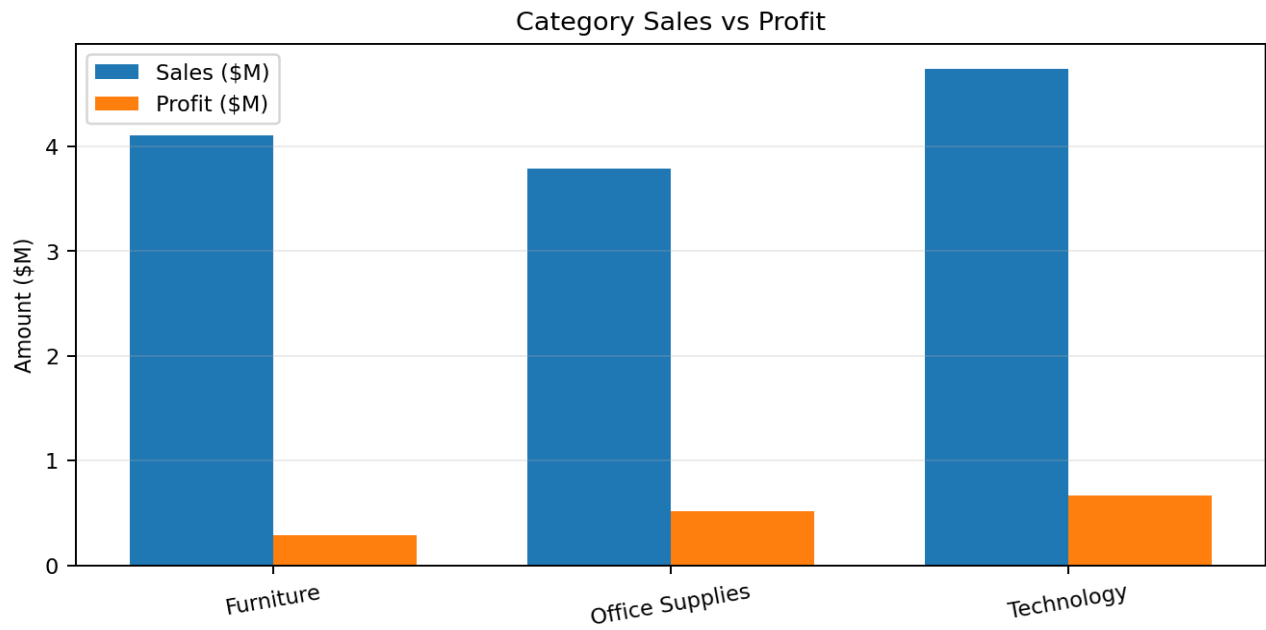
Annual performance shows consistent growth. Sales increased each year from \$2.26M in 2011 to \$4.30M in 2014. Profit rose from \$248.9K to \$504.2K. The highest annual sales and profit occurred in 2014.



Year	Sales	Profit	Margin	Orders
2011	\$2.26M	\$248.9K	11.02%	4,440
2012	\$2.68M	\$307.4K	11.48%	5,343
2013	\$3.41M	\$406.9K	11.95%	6,721
2014	\$4.30M	\$504.2K	11.73%	8,531

7. Category Performance

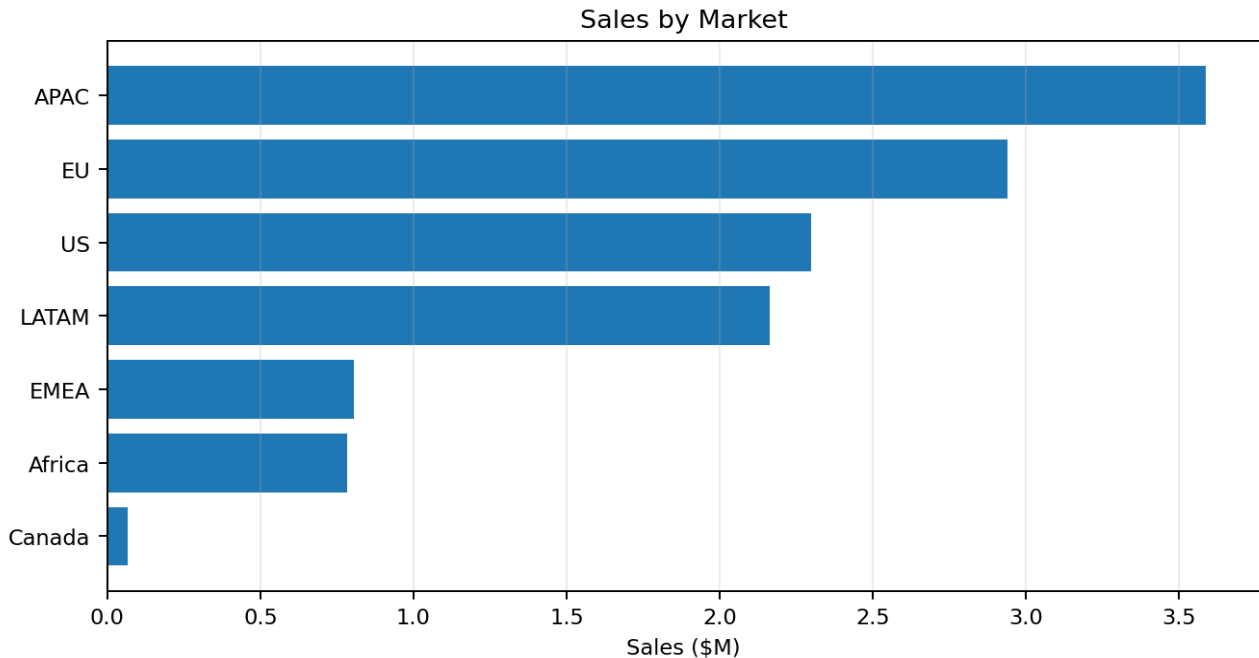
Technology is the leading category by sales at \$4.74M and also has the highest category margin at 13.99%. Office Supplies generates \$3.79M with a 13.69% margin. Furniture generates \$4.11M, but its margin is materially lower at 6.94%, indicating a need for stronger pricing and cost controls.



Category	Sales	Profit	Margin	Quantity
Furniture	\$4.11M	\$285.2K	6.94%	34,954
Office Supplies	\$3.79M	\$518.5K	13.69%	108,182
Technology	\$4.74M	\$663.8K	13.99%	35,176

8. Market & Geographic Analysis

APAC is the largest market with \$3.59M in sales and \$436.0K in profit. EU follows with \$2.94M. The US contributes \$2.30M. EMEA is comparatively weak on margin at only 5.45%, suggesting opportunities for cost, pricing, or product-mix optimization.



Market	Sales	Profit	Margin
APAC	\$3.59M	\$436.0K	12.16%
EU	\$2.94M	\$372.8K	12.69%
US	\$2.30M	\$286.4K	12.47%
LATAM	\$2.16M	\$221.6K	10.24%
EMEA	\$806.2K	\$43.9K	5.45%
Africa	\$783.8K	\$88.9K	11.34%
Canada	\$66.9K	\$17.8K	26.62%

Top countries by sales: United States (\$2.30M), Australia (\$925.2K), France (\$858.9K), China (\$700.6K), and Germany (\$628.8K).

Top states/regions by sales:

State	Sales	Profit
England	\$485.2K	\$99.9K
California	\$457.7K	\$76.4K
Ile-de-France	\$317.8K	\$44.1K
New York	\$310.9K	\$74.0K
New South Wales	\$270.5K	\$43.7K
Queensland	\$238.3K	\$21.6K
North Rhine-Westphalia	\$216.5K	\$42.3K

State	Sales	Profit
Texas	\$170.2K	\$-25.7K
San Salvador	\$153.6K	\$35.9K
National Capital	\$152.2K	\$-13.1K

9. Customer Segment & Shipping Analysis

The Consumer segment is the largest contributor with \$6.51M in sales and \$749.2K in profit. Corporate contributes \$3.82M, while Home Office contributes \$2.31M.

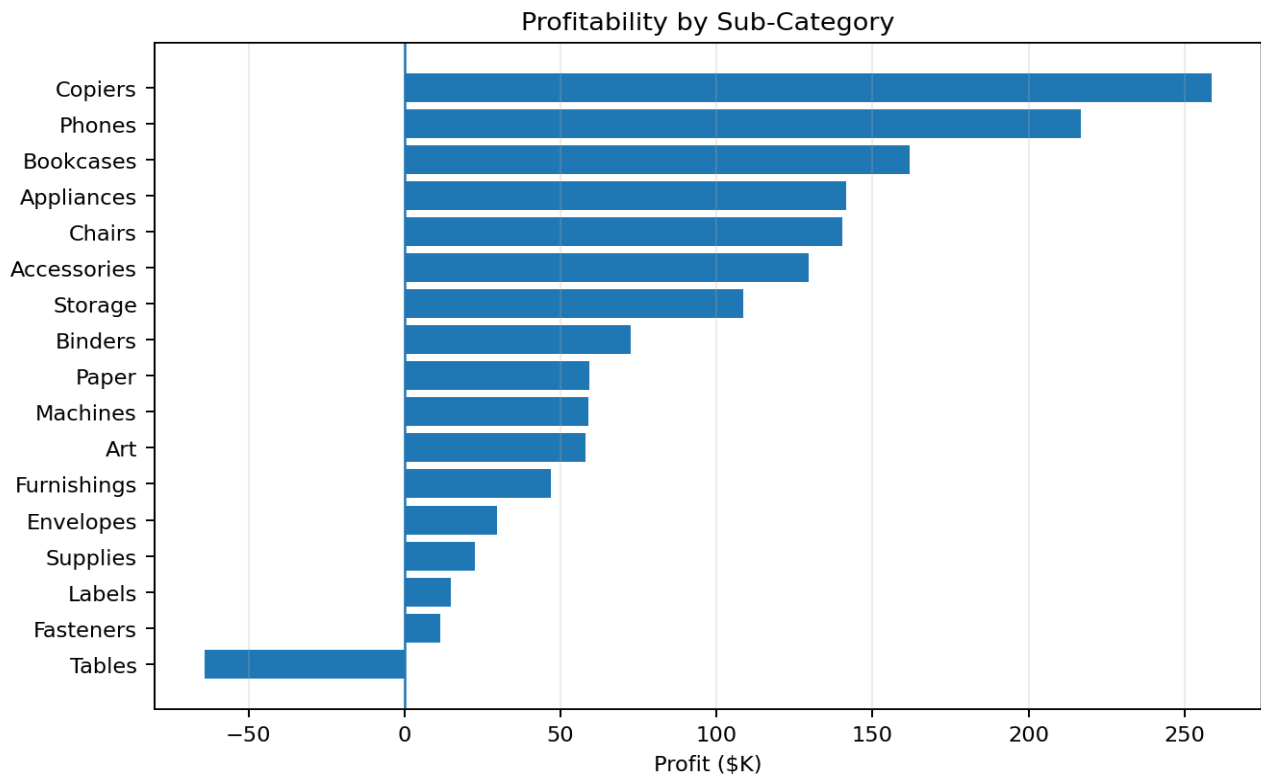
Segment	Sales	Profit	Margin	Orders
Consumer	\$6.51M	\$749.2K	11.51%	13,104
Corporate	\$3.82M	\$441.2K	11.54%	7,673
Home Office	\$2.31M	\$277.0K	11.99%	4,687

Shipping: Standard Class dominates with \$7.58M in sales and \$890.6K in profit. Second Class contributes \$2.57M; First Class contributes \$1.83M; Same Day contributes \$667.2K.

Ship Mode	Sales	Profit	Orders
Standard Class	\$7.58M	\$890.6K	15,154
Second Class	\$2.57M	\$292.6K	5,119
First Class	\$1.83M	\$208.1K	3,821
Same Day	\$667.2K	\$76.2K	1,347

10. Product & Sub-Category Profitability

Phones is the top sub-category by sales at \$1.71M, followed by Copiers and Chairs. However, profitability is not always aligned with revenue. Tables is the key loss-making sub-category with \$-64.1K in negative profit. This is a critical business finding because continued revenue growth in a loss-making product line can reduce overall profitability.



Sub-Category	Sales	Profit	Margin
Tables	\$757.0K	\$-64.1K	-8.46%
Fasteners	\$83.2K	\$11.5K	13.85%
Labels	\$73.4K	\$15.0K	20.45%
Supplies	\$243.1K	\$22.6K	9.29%
Envelopes	\$170.9K	\$29.6K	17.32%
Furnishings	\$385.6K	\$47.0K	12.18%
Art	\$372.1K	\$58.0K	15.58%
Machines	\$779.1K	\$58.9K	7.56%

Risk signal: The dataset contains many negative-profit transaction lines. This suggests that discounting, product cost, shipping cost, or pricing strategy should be investigated at transaction and product level rather than relying only on sales volume.

11. Business Problems Identified

- Furniture has a relatively low margin (6.94%) compared with Technology and Office Supplies.
- Tables is loss-making at \$-64.1K.
- EMEA has the lowest market margin at 5.45%.

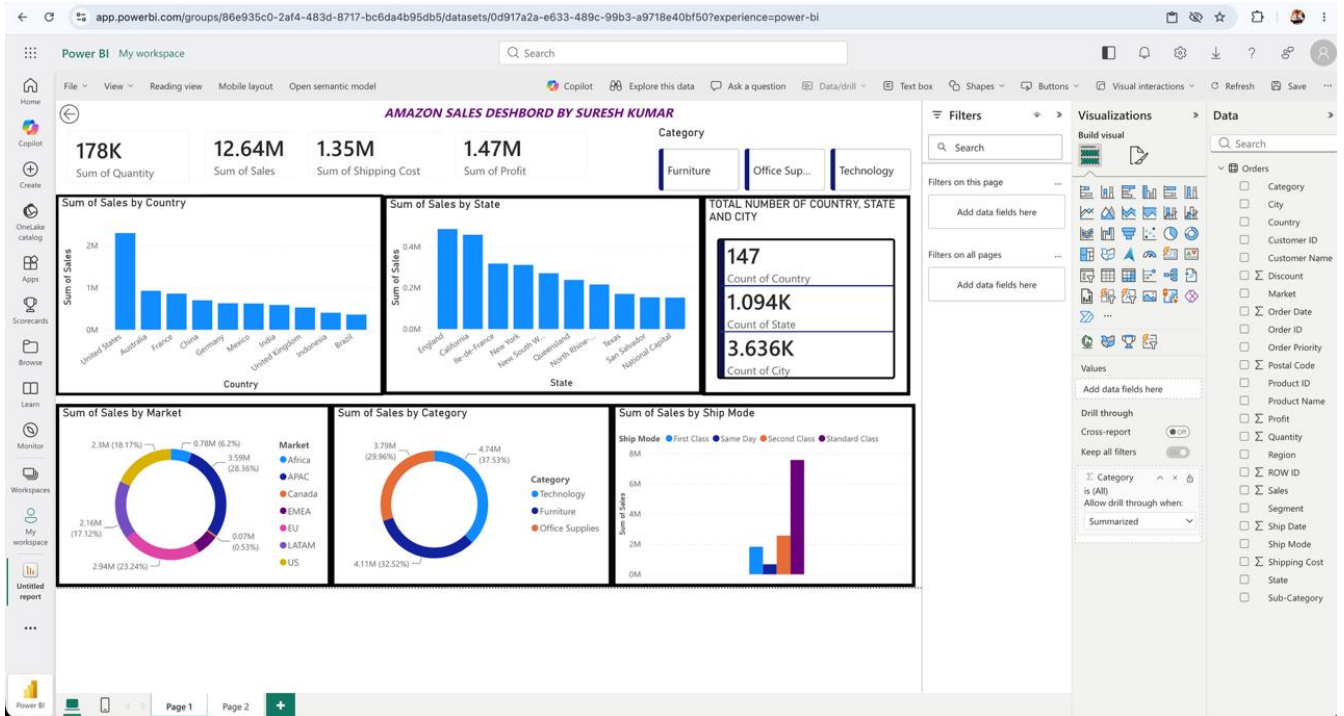
- The weighted average discount is 0.11%, requiring monitoring for margin erosion.
- Return rate is approximately 4.69% of distinct orders, which creates a measurable opportunity for product and service quality improvement.
- Some geographic areas show negative profitability despite generating sales, indicating a need for localized pricing and cost analysis.

12. Business Recommendations

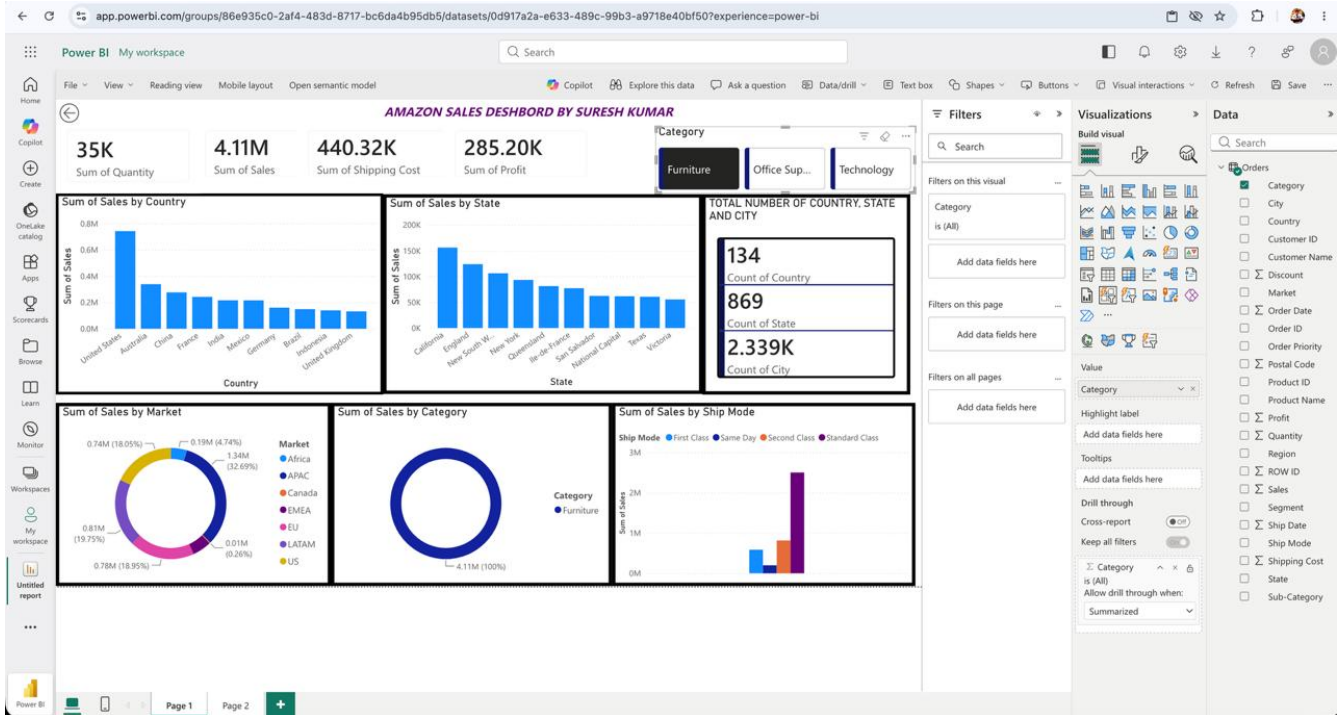
- Optimize Furniture pricing: Review pricing, supplier costs, freight, and discount levels for Furniture—especially Tables.
- Control discounting: Introduce margin-based discount thresholds so discounts are approved only when the expected contribution remains positive.
- Prioritize high-margin growth: Expand Technology and Office Supplies where revenue and profitability are both strong.
- Improve EMEA economics: Analyze freight, product mix, and discount behavior in EMEA to close the margin gap.
- Investigate returns: Analyze returned products by category, market, product, and customer segment to identify recurring quality or fulfillment issues.
- Use regional pricing: Avoid a one-size-fits-all pricing strategy. Use market-level profitability and logistics costs to guide localized decisions.
- Monitor shipping economics: Standard Class is the dominant mode, so improving its fulfillment efficiency can have a large financial impact.
- Build management alerts: Add Power BI alerts for negative-profit products, declining margins, high returns, and unusual discount levels.

13. Power BI Dashboard

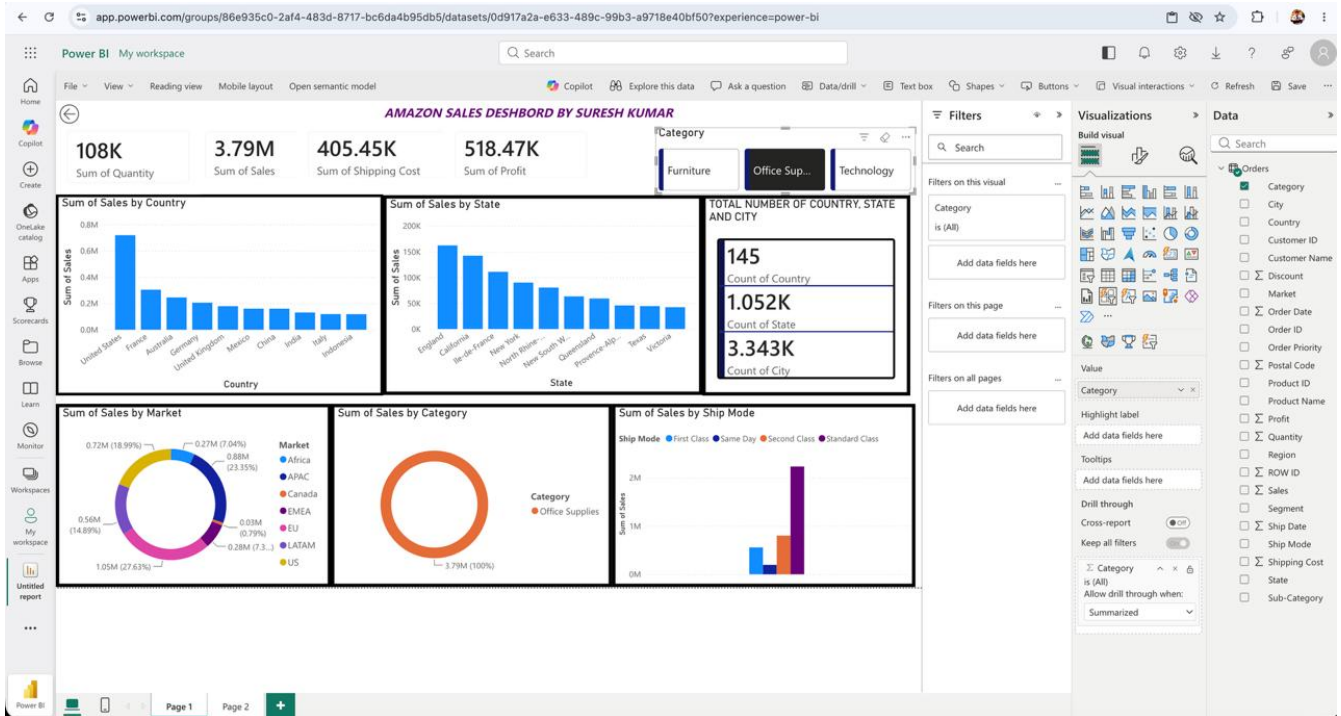
The dashboard is designed for interactive business analysis. KPI cards summarize quantity, sales, shipping cost, and profit. Country and state charts provide geographic performance, while donut charts show market/category contribution. The Category slicer allows management to compare Furniture, Office Supplies, and Technology without rebuilding the report.



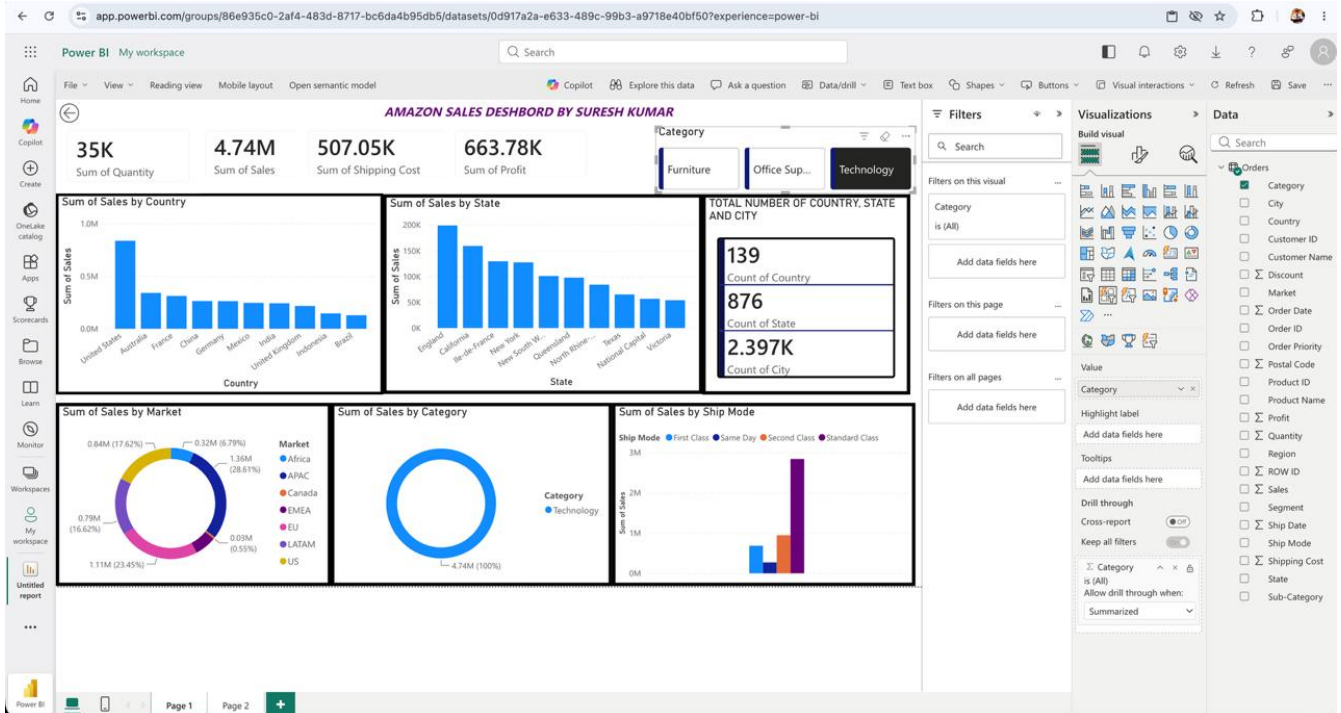
Dashboard - All Categories



Dashboard - Office Supplies



Dashboard - Furniture



Dashboard - Technology

14. Key Portfolio Takeaways

- Demonstrated end-to-end data analytics from Excel ingestion to Power BI dashboarding.
- Analyzed 51,290 order records across 147 countries and 7 markets.
- Produced business KPIs totaling \$12.64M sales and \$1.47M profit.
- Identified high-value categories, markets, customer segments, and shipping modes.
- Detected a material profitability issue in Tables (\$-64.1K profit).
- Quantified a 4.69% return rate and 0.11% weighted discount.

- Converted analytical results into pricing, product, logistics, and market recommendations.

15. Conclusion

This project demonstrates how data analytics and business analytics can transform raw e-commerce transactions into decision-ready intelligence. The Power BI dashboard enables users to move from high-level KPIs to category, geographic, market, and shipping-level insights. The most important management opportunity is to protect margin while continuing growth—particularly by improving Furniture and Tables profitability, controlling discounts, optimizing weaker markets, and using return and shipping data to improve operational efficiency.

Recommended next phase: add a monthly sales trend page, customer lifetime-value analysis, product-level Pareto analysis, return reasons, and a profitability alert dashboard.

Prepared for Data Analyst / Business Analyst portfolio presentation.